



Luis E. Valentin-Alvarado, Ph.D.

Postdoctoral Scholar · Jennifer A. Doudna Lab
Innovative Genomics Institute · University of California, Berkeley
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Research Interests

My research connects ecosystem-scale genomics with enzyme function to understand how bacteria, archaea, and mobile genetic elements shape microbial communities. I am particularly interested in combining AI for biology with structural bioinformatics to uncover uncharacterized enzyme functions and understand RNA-guided systems, gene regulation, and prokaryotic defense. By integrating genomics and evolutionary analysis with biochemistry and structural biology, I aim to connect computational predictions with experimental evidence to reveal molecular mechanisms—from carbon and energy metabolism in wetland soils and biofilms to nucleic acid signaling in microbial defense.

Academic and Research Appointments

Postdoctoral Scholar, Doudna Lab, UC Berkeley Sep 2026–present

Investigating tRNA-derived second messengers and the origins, processing, and fates of nucleic acid signals in prokaryotic defense systems through biochemistry, bacterial genetics, and structural biology.

Group Leader, Monash University Nov 2025–Aug 2026

Department of Biochemistry and Molecular Biology. Directed research priorities and collaborations; supervised students and junior researchers across computational and experimental projects.

Visiting Researcher, Doudna Lab, UC Berkeley Sep 2025–Aug 2026

Innovative Genomics Institute. Led collaborative research on Cas13-associated defense systems using RNA biochemistry, bacterial genetics, fluorescence assays, and microscopy.

Research Fellow, Knott Lab, Monash University Aug 2024–Oct 2025

Led discovery and characterization of CRISPR-Cas systems and accessory proteins through comparative genomics, protein expression, RNA biochemistry, and bacterial assays.

Postdoctoral Fellow, UC Berkeley Mar–Jul 2024

Adjunct Instructor, Contra Costa College Aug 2022–present

Research Assistant, Woods Hole Oceanographic Institution and MIT 2016–2018

Education and Advanced Training

PhD, Microbiology · University of California, Berkeley 2018–2024

Research mentors: Jillian F. Banfield and David F. Savage. Dissertation: “Microbial Tapestry: Interwoven activities of novel uncultivated microorganisms, their genome structures, and mobile genetic elements in subsurface ecosystems.”

BSc, Chemistry and Microbiology · University of Puerto Rico, Humacao 2011–2016

Magna cum laude; minor in Curriculum Development and Pedagogy.

Microbial Diversity Summer Course, Marine Biological Laboratory, Woods Hole · 2022

Fellowships and Awards

2025 · Human Frontier Science Program Long-Term Fellowship. Project: “Systematic discovery of evolutionary origins and functional diversity of methyltransferases.”

2023 · Best Graduate Student Oral Presentation, SACNAS; UC Berkeley Dissertation Fellowship.

2022 · Research University Alliance research exchange at Harvard; UC Berkeley Graduate Division Travel Award; MBL Microbial Diversity Course Scholarship.

Publications

Name shown in bold; † denotes equal contribution. Ellipses indicate omitted coauthors.

Research Articles

1. Yoon PH, Docter TA, Zhang ZT, Loi K, Lopez SC, **Valentin-Alvarado LE**, Tuck OT, Brohawn SG, Doudna JA (2026). VIPR RNA-guided DNA recognition by noncontiguous geometric triplex formation. *Science* 393(6817):1236–1240. doi:10.1126/science.aei3472
2. Wirawan R, Jamieson WD, Baird HM, Berry C, Lupton CJ, Venugopal H, **Valentin-Alvarado LE**, Best HL, Jones DD, Williamson LJ, Auhim HS, Castell OK, Dunstone MA, Spicer BA (2026). Structural basis for independent pore function of Vpb4 from *Bacillus thuringiensis*. *Nature Communications* 17:8198. doi:10.1038/s41467-026-74567-y
3. Hamm JN, Dombrowski N, **Valentin-Alvarado LE**, Greening C, Williams TA, Spang A (2026). New lineages provide insights into the convergent evolution of extreme salt adaptation within symbiotic Archaea. *Molecular Biology and Evolution* 43(5):msag091. doi:10.1093/molbev/msag091
4. Banfield JF, **Valentin-Alvarado LE**, Shi L-D, Robinson CM, Bamert RS, Coulibaly F, Barth ZK, Aylward FO, Schoelmerich MC, Lei S, Sachdeva R, Knott GJ (2025). Convergent evolution of viral-like Borg archaeal extrachromosomal elements and giant eukaryotic viruses. *Nature Communications* 16:10641. doi:10.1038/s41467-025-65646-7
5. Amano Y, Sachdeva R, Gittins D, Anantharaman K, Lei S, **Valentin-Alvarado LE**, Diamond S, Beppu H, Iwatsuki T, Mochizuki A, Miyakawa K, Ishii E, Murakami H, Jaffe AL, Castelle C, Lavy A, Suzuki Y, Banfield JF (2024). Diverse microbiome functions, limited temporal variation and substantial genomic conservation within sedimentary and granite rock deep underground research laboratories. *Environmental Microbiome* 19:105. doi:10.1186/s40793-024-00649-3
6. **Valentin-Alvarado LE**, Shi L-D, Appler KE, Crits-Christoph A, De Anda V, Adler BA, Cui ML, Ly L, Leão P, Roberts RJ, Sachdeva R, Baker BJ, Savage DF, Banfield JF (2024). Complete genomes of Asgard archaea reveal diverse integrated and mobile genetic elements. *Genome Research* 34(10):1595–1609. doi:10.1101/gr.279480.124
Contribution: Led the study and developed an experimental–computational workflow to analyze archaeal DNA methylation with PacBio HiFi data and link methylation motifs to mobile genetic elements.
7. **Valentin-Alvarado LE†**, Appler KE†, De Anda V, Schoelmerich MC, West-Roberts J, Kivenson V, Crits-Christoph A, Ly L, Sachdeva R, Greening C, Savage DF, Baker BJ, Banfield JF (2024). Asgard archaea modulate potential methanogenesis substrates in wetland soil. *Nature Communications* 15:6384. doi:10.1038/s41467-024-49872-z
Contribution: Co-led the study, including long-read genome reconstruction and comparative metabolic and phylogenetic analyses of soil Asgard archaea.

Publications (continued)

Research Articles (continued)

8. Greening C†, Cabotaje PR†, **Valentin Alvarado LE**†, Leung PM†, Land H, Rodrigues-Oliveira T, Ponce-Toledo RI, Senger M, Klamke MA, Milton M, Lappan R, Mullen S, West-Roberts J, Mao J, Song J, Schoelmerich M, Stairs CW, Schleper C, Grinter R, Spang A, Banfield JF, Berggren G (2024). Minimal and hybrid hydrogenases are active from archaea. *Cell* 187(13):3357–3372.e19. doi:10.1016/j.cell.2024.05.032
Contribution: Co-first author; contributed metagenomic datasets and genomes, homology searches, genome-wide metabolic reconstruction, and phylogenetic analyses of archaeal hydrogenases.
9. **Valentin-Alvarado LE**, Fakra SC, Probst AJ, Giska JR, Jaffe AL, Oltrogge LM, West-Roberts J, Rowland J, Manga M, Savage DF, Greening C, Baker BJ, Banfield JF (2024). Autotrophic biofilms sustained by deeply sourced groundwater host diverse bacteria implicated in sulfur and hydrogen metabolism. *Microbiome* 12:15. doi:10.1186/s40168-023-01704-w
Contribution: Led metagenome-assembled genome recovery and metabolic reconstruction linking subsurface biofilm communities to inferred sulfur and hydrogen cycling.
10. Al-Shayeb B, Schoelmerich MC, West-Roberts J, **Valentin-Alvarado LE**, Sachdeva R, Mullen S, Crits-Christoph A, Wilkins MJ, Williams KH, Doudna JA, Banfield JF (2022). Borgs are giant genetic elements with potential to expand metabolic capacity. *Nature* 610:731–736. doi:10.1038/s41586-022-05256-1
11. Crits-Christoph A, Diamond S, Al-Shayeb B, **Valentin-Alvarado L**, Banfield JF (2022). A widely distributed genus of soil Acidobacteria genomically enriched in biosynthetic gene clusters. *ISME Communications* 2:70. doi:10.1038/s43705-022-00140-5
12. Jaffe AL, Thomas AD, He C, Keren R, **Valentin-Alvarado LE**, Munk P, Bouma-Gregson K, Farag IF, Amano Y, Sachdeva R, West PT, Banfield JF (2021). Patterns of gene content and co-occurrence constrain the evolutionary path toward animal association in Candidate Phyla Radiation bacteria. *mBio* 12(4):e00521-21. doi:10.1128/mBio.00521-21
13. Ehrenberg AJ, Moehle EA, Brook CE, ... **Valentin-Alvarado L**, ... Doudna JA; IGI SARS-CoV-2 Testing Consortium (2021). Launching a saliva-based SARS-CoV-2 surveillance testing program on a university campus. *PLoS ONE* 16(5):e0251296. doi:10.1371/journal.pone.0251296
14. Nichols RJ, LaFrance B, Phillips NR, Radford DR, Oltrogge LM, **Valentin-Alvarado LE**, Bischoff AJ, Nogales E, Savage DF (2021). Discovery and characterization of a novel family of prokaryotic nanocompartments involved in sulfur metabolism. *eLife* 10:e59288. doi:10.7554/eLife.59288
15. Hawco NJ, McIlvin MM, Bundy RM, Tagliabue A, Goepfert TJ, Moran DM, **Valentin-Alvarado L**, DiTullio GR, Saito MA (2020). Minimal cobalt metabolism in the marine cyanobacterium Prochlorococcus. *Proceedings of the National Academy of Sciences* 117(27):15740–15747. doi:10.1073/pnas.2001393117

Publications (continued)

Reviews, Perspectives, Editorials, and Education Scholarship

16. Van Den Bossche T, Lazau EA, Aho VTE, Blakeley-Ruiz JA, Wolf M, Kunath BJ, **Valentin-Alvarado LE**, ... Jagtap PD (2026). Integrating multi-omics technologies to decipher microbiome functions. *Nature Communications* 17:9600. doi:10.1038/s41467-026-77539-4
17. **Valentin-Alvarado LE**, Knott GJ (2025). From code to comprehension: AI captures the language of life. *The CRISPR Journal* 8(1):2–4. doi:10.1089/crispr.2025.0008
18. Prywes N, Phillips NR, Tuck OT, **Valentin-Alvarado LE**, Savage DF (2023). Rubisco function, evolution, and engineering. *Annual Review of Biochemistry* 92:385–410. doi:10.1146/annurev-biochem-040320-101244
19. Tully BJ, Buongiorno J, Cohen AB, Cram JA, Garber AI, Hu SK, Krinos AI, Leftwich PT, Marshall AJ, Sieradzki ET, Speth DR, Suter EA, Trivedi CB, **Valentin-Alvarado LE**, Weissman JL; BVCN Instructor Consortium (2021). The Bioinformatics Virtual Coordination Network: An open-source and interactive learning environment. *Frontiers in Education* 6:711618. doi:10.3389/feduc.2021.711618
20. Scott Price O, Luis K, Price AL, Harden B, Howard J, **Valentin LE**, Liles G, Jearld A Jr (2020). Same program different delivery: Adapting the Woods Hole Partnership Education Program for a virtual era. *Limnology and Oceanography Bulletin* 29(4):117–124. doi:10.1002/lob.10414
21. Baltar F, Bayer B, Bednarsek N, Deppeler S, Escribano R, Gonzalez CE, Hansman RL, Mishra RK, Moran MA, Repeta DJ, Robinson C, Sintes E, Tamburini C, **Valentin LE**, Herndl GJ (2019). Towards integrating evolution, metabolism, and climate change studies of marine ecosystems. *Trends in Ecology & Evolution* 34(11):1022–1033. doi:10.1016/j.tree.2019.07.003

Preprints

- P1. Wang S, Guiton AK, **Valentin-Alvarado LE**, Garner R, Zhang P, Yan M, Shi L-D, Schoelmerich MC, Steininger HM, Portik DM, Zhang S, Wilkinson JE, Lynch S, Morowitz MJ, Hess M, Diamond S, Banfield JF, Sachdeva R (2026). Metagenomic strain-resolved DNA modification patterns link extrachromosomal genetic elements to host strains. *bioRxiv*. doi:10.64898/2026.03.27.714056
- P2. West-Roberts JA†, **Valentin-Alvarado LE†**, Mullen S, Sachdeva R, Smith J, Hug LA, Gregoire DS, Liu W, Lin TY, Husain G, Amano Y, Ly L, Banfield JF (2023). Giant genes are rare but implicated in cell wall degradation by predatory bacteria. *bioRxiv*. doi:10.1101/2023.11.21.568195

Manuscript in Preparation

Valentin-Alvarado LE, Sachdeva R, Savage DF, Banfield JF. Wetland soil hosts a high diversity of novel archaea actively expressing pathways for carbon cycling and represented by complete genomes.

Teaching Experience

Instructor, General Microbiology · Contra Costa College 2022–present

Appointment held continuously since 2022; returned to teaching after moving back to Berkeley. Design and deliver undergraduate course content connecting microbiology to health, environmental science, and biotechnology. Develop interactive lessons and mentor students from underrepresented backgrounds in STEM.

Graduate Student Instructor · UC Berkeley 2021

Principles of Biochemistry, Department of Molecular and Cell Biology. Led discussion sections and office hours; developed and graded problem sets for a course serving graduate and undergraduate students.

Graduate Student Instructor · UC Berkeley 2020

General Microbiology Laboratory, Department of Plant and Microbial Biology. Led discussions, assigned and graded essays, and delivered a guest lecture on metagenomics.

Guest Lecturer, CRISPR Biology · IGI / UC Berkeley 2020

Undergraduate Teaching Assistant · University of Puerto Rico 2016

Molecular Cell Biology Laboratory. Developed active laboratory assignments in microbiology, microbial ecology, and molecular biology; supported students during laboratory work.

Supervision and Research Mentoring

Doctoral co-supervision (2025–Aug 2026): Natasha Jablonsky, Monash University.

Doctoral advisory panel (2025–Aug 2026): Warunika Rankiri Pathirannahelage, Monash University.

Graduate research mentoring: Saoirse Disney-McKeethen, UC Berkeley · 2024.

Undergraduate research mentoring (2022–2024): Michael Li Cui and Astrid Adrianna Quirarte, UC Berkeley.

Earlier Research Training

Research Assistant · Woods Hole Oceanographic Institution and MIT 2016–2018

Mak Saito and Otto Cordero laboratories. Developed targeted metaproteomics and protein-production methods; investigated marine microbial metalloenzymes and metal-binding ligands using molecular biology and mass spectrometry.

Undergraduate Researcher · University of Georgia Summers 2014, 2015

Jorge Escalante-Semerena laboratory. Studied PocR-associated physiology and metabolism in *Salmonella* using molecular genetics, protein purification, and biochemical assays.

Undergraduate Researcher · University of Puerto Rico 2014, 2016

Lilliam Casillas-Martinez laboratory. Developed computational approaches to characterize repetitive satellite DNA sequences.

Undergraduate Researcher · University of Puerto Rico 2012–2014

Margarita Ortiz-Marciales laboratory. Investigated asymmetric synthesis using boron compounds for chirality transfer and preparation of chiral molecules.

Research Methods

Genome-resolved metagenomics; long-read genome assembly (PacBio and Nanopore); comparative genomics and phylogenomics; DNA methylation analysis; Python, R, and Bash; RNA biochemistry; bacterial genetics; recombinant protein expression and purification; fluorescence assays and microscopy.

Invited Talks and Conference Presentations

Scheduled Invited Seminar

October 5, 2026 · “Evolutionary conflicts and signaling in microbial defense systems.” Extreme Biology Institute, Montana State University; virtual seminar.

Invited Talks and Seminars

November 20, 2025 · “Complete genomes from complex soils: Long-Read PacBio metagenomics for high-resolution microbial ecology.” APAC Microbiome Leadership Forum, Seoul, South Korea. Program

October 2024 · “High-fidelity long reads enable hundreds of contiguous MAGs in wetland soil microbiome.” PacBio metagenomics webinar; broadcasts October 1 and 9. Recording

October 1, 2024 · “High-fidelity long reads enable hundreds of contiguous MAGs in wetland soil microbiome.” ASM Hour, Australian Society for Microbiology; Equity, Diversity and Inclusion session.

May 14, 2024 · PacBio PRISM, San Francisco, California; invited presentation.

2024 · 19th International Symposium on Microbial Ecology (ISME19), Cape Town, South Africa.

2024 · Biomolecular Horizons, Melbourne, Australia.

2024 · International Conference on Geo-Omics of Archaea, Shenzhen, China.

2024 · Deciphering Microbiome Functions, Lorentz Center, Netherlands.

2022 · Microbiology Department, University of Puerto Rico, Mayagüez.

2022 · Chemical Biology, Harvard University; host: Emily Balskus.

Additional Presentations

June 18, 2025 · “High-fidelity long-read sequencing enables recovery of hundreds of contiguous MAGs from wetland soils.” JAMS Brisbane, PacBio Special, Brisbane, Australia.

2024 · Geobiology Gordon Research Conference, Galveston, Texas.

2024 · 29th Molecular Microbial Ecology Group Meeting, London, United Kingdom.

March 19, 2024 · PhD finishing seminar, UC Berkeley.

March 30, 2023 · “Enigmatic mobile genetic elements in complete Asgard archaeal genomes.” Student and Postdoc Seminar, UC Berkeley.

2022 · International Symposium on Microbial Ecology (ISME), Lausanne, Switzerland; oral presentation.

2022 · Microbial Community Ecology, Max Planck Institute, Plön, Germany; oral presentation.

2021 · Plant and Microbial Biology Graduate Student Seminar, UC Berkeley.

Public Engagement and Outreach

August 2024 · **Panelist, Diversity in STEM**, ISME19, Cape Town, South Africa.

December 12, 2020 · **Keynote speaker and Spanish breakout leader**, TeenTechSF Global Youth Summit, “Tech & Health: Understand More, Fear Less.”

October 21, 2020 · **Panelist, Carreras en Ciencias**, Día de la Ciencia, Bay Area Science Festival. Science at Cal and the Consulate General of Mexico in San Francisco; Spanish-language public event. Event

September 16, 2017 · **Co-organizer, OCEANOS**, Woods Hole Oceanographic Institution. Bilingual public-science symposium in Spanish and Portuguese. Event

Science Communication and Media

Videos and Recorded Interviews

2022 · Featured scientist and panelist, Un Vistazo al Laboratorio. Science at Cal, UC Berkeley, and the Consulate General of Mexico in San Francisco. Spanish-language laboratory tour and discussion of scientific careers. Videos · YouTube playlist

March 24, 2021 · Interview guest, BACR News / Lawrence Hall of Science. Student-led recorded interview; featured in the Lawrence Hall of Science Community Connections exhibit in 2022.

November 2020 · Featured scientist, CRISPR Lights Up COVID-19 Testing. Science at Cal and the Lawrence Hall of Science. Educational videos in English and Spanish. English video · Spanish video

Selected Interviews and Research Profiles

May 3, 2025 · Primera Hora. “De Santa Isabel al mundo: Boricua hace historia con prestigiosa beca científica en Australia.” Interview by Sara R. Marrero Cabán. Article

September 8, 2024 · El Nuevo Día. “Santa Isabel scientist discovers microorganism and names it in honor of the Taíno goddess Atabey.” Profile by Valeria María Torres Nieves. Article

August 7, 2024 · UC Berkeley Plant and Microbial Biology. “Shaping our understanding of microbial communities.” Research and career interview. Interview

Spring 2022 · Breakthroughs, UC Berkeley Rausser College. “Deciphering Microbial Mysteries.” Research profile by Zac Unger. Article

Professional and University Service

Journal peer review (2024–2025): The ISME Journal; Nature Microbiology; Environmental Microbiology; Microbiome; Nature Communications; Communications Biology; Communications Earth & Environment.

2019–2024 · Graduate Student Director, Cal NERDS. Supported undergraduate PhD admissions, fellowship applications, and retention in STEM.

2020–2022 · Committee Member, Diversity, Equity, and Inclusion, Innovative Genomics Institute.

2019–2020 · Program Coordinator, STEM First-Year Initiative, Graduate Office for Diversity, UC Berkeley.

2019–2020 · Committee Member, Science Advocacy, Communication, and Outreach, UC Berkeley–UCSF Bioengineering PhD program.

2019–2020 · PhD Admissions Mentor, Científico Latino.

2019–2020 · Graduate Student Mentor, Out in STEM; mentored two graduate students.

2018–2019 · Student Chair, Plant and Microbial Biology Department Seminars, UC Berkeley.

2018–2019 · Fellowship Application Mentor, Latino Association of Graduate Students in Engineering and Science, UC Berkeley.

Patent Application

Doudna J, Banfield J, Shivram H, Tsuchida CA, Al-Shayeb B, Sachdeva R, Jaffe A, **Valentin-Alvarado L.** “Modulators of Type II-A CRISPR-Cas effector polypeptides and methods for the use thereof.” Application filed.